

Tests & Quizzes

Question Pool: Quiz 03

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Question 1: Single Correct - 1.0 point

If you receive bunch of gibberish sent by Alice and you decide to check the authenticity of the message by checking the digital signature, and the digital signature checks out as authentic but you still doubt the message contents then which one of these statements is true:

- I. Either Alice's private key is compromised
- II. Alice has sent the message by mistake
- III. Some attacker intercepted the message, and changed the contents before sending it to you
- IV. Some attacker completely made their own new message and sent it under Alice's name

- ☐ OA. I, II III
- ☐ OB. I, II III
- ☐ OC. All of them
- ☒ OD. I, II, IV

Answer Key: D

Question 2: Single Correct - 1.0 point

A protocol change results in upgraded nodes rejecting blocks that legacy nodes still accept, while legacy nodes accept all blocks produced by upgraded nodes.
What property is being demonstrated?

- ☐ OA. Forward compatibility
- ☒ OB. Backward compatibility
- ☐ OC. Consensus irreversibility
- ☐ OD. State finality

Answer Key: B

Question 3: Single Correct - 1.0 point

Which statement best explains cost-push inflation?

- ☐ OA. Firms raise prices because consumers are willing to pay more.
- ☒ OB. Rising wages or input costs lead producers to increase prices.
- ☐ OC. Increased government spending raises aggregate demand.
- ☐ OD. Excess liquidity in financial markets raises asset prices.

Answer Key: B

Question 4: Single Correct - 1.0 point

What is the primary distinction between digital cash and e-cash in most cryptographic payment models?

- ☐ OA. Digital cash relies on centralized databases, whereas e-cash operates only on decentralized networks.
- ☒ B. Digital cash represents electronic monetary value broadly, while e-cash is specifically designed to provide cash-like anonymity through cryptographic techniques.
- ☐ OC. Digital cash requires physical backing by reserves, whereas e-cash is purely algorithmic.
- ☐ OD. Digital cash supports peer-to-peer transfers, whereas e-cash is limited to account-based transactions.

Answer Key: B

Question 5: Single Correct - 1.0 point

Suppose you want to trade using Bitcoin while also leveraging smart contract functionality. Which option is best suited for this use case?

- ☐ OA. Executing transactions directly on the Bitcoin base layer using its native scripting capabilities
- ☒ B. Using a Layer-2 or sidechain solution like wBTC that enables programmable smart contract execution
- ☐ OC. Migrating funds to an independent smart contract platform like Ethereum with a separate native token model
- ☐ OD. Conducting trades through a centralized platform like Binance offering automated conditional order features

Answer Key: B

Question 6: Single Correct - 1.0 point

Why is detecting double spending generally easier than completely preventing it in digital payment systems?

- ☒ A. Because detecting double spending only requires identifying conflicting uses of the same digital coin, whereas preventing it requires coordination to stop acceptance beforehand.
- ☐ OB. Because detecting double spending relies mainly on encryption techniques, while preventing it depends on digital signatures and identity verification.
- ☐ OC. Because detecting double spending can be performed by observing transaction history, whereas preventing it depends on limiting the number of transactions users can make.
- ☐ OD. Because detecting double spending focuses on reviewing past activity, while preventing it requires increasing computational difficulty for attackers.

Answer Key: A

Question 7: Single Correct - 1.0 point

In Chaumian e-cash, why is user anonymity preserved even though the issuing bank must verify coin validity at redemption?

- ☐ A. Because the bank stores only hashed transaction records
- ☒ B. Because the coin's serial number is blinded during signing and later revealed without identity linkage
- ☐ C. Because consensus nodes collectively validate transactions
- ☐ D. Because the coin is encrypted using the user's public key

Answer Key: B

Question 8: Single Correct - 1.0 point

Which statement best describes smart contracts in the context of blockchain systems?

- ☒ A. Smart contracts are automated computer programs stored on a blockchain that execute predefined rules or agreements when specified conditions are met without requiring manual intervention.
- ☐ B. Smart contracts are digital documents used to record legal agreements between parties, which are stored securely on centralized servers for future reference and auditing.
- ☐ C. Smart contracts are encrypted communication protocols that allow users to securely exchange messages and financial data across distributed computer networks.
- ☐ D. Smart contracts are consensus algorithms that enable network participants to agree on the validity of transactions and maintain synchronization across nodes.

Answer Key: A

Question 9: Single Correct - 1.0 point

What is a key benefit of tokenizing stocks?

- ☒ A. Enables fractional ownership for investors
- ☐ B. Keeps stock prices stable over time
- ☐ C. Removes regulatory oversight requirements
- ☐ D. Guarantees instant settlement of trades

Answer Key: A

Return